Arthur Cayley

Collected Mathematical Papers

Volume IX, Pages 576–600

Transcribed from the Original Publications

Contents

623. On Three-Bar Motion (continued)

[Continued from earlier pages of the same paper.]

[p. 576] Single-infinite series of ∞ and circumscribed triangles, so that, drawing from a point A of the circle tangents to the parabola again meeting the circle in the points B and C respectively, BC will be a tangent to the parabola; this is in fact a known thing, but, repeating the same kind of investigation as before, we can, with the affinity points K on the circle, the parabola described to touch two of the sides of the triangle will touch the third side.

Taking, then, a circle radius $\frac{1}{2}k$, and upon it the three points A, B, C determined by the angles 2α, 2β, 2γ respectively (viz. the coordinates of A are $x = \frac{1}{2}k\cos 2\alpha$, $\frac{1}{2}k\sin 2\alpha$, &c.), and a point K determined by the angle 2ω (suppose for a moment $\omega = 0$), then the lines from K, viz. the lines having for K the equations &c. at the foot of A, and the lines KB with the fixed parabola, are connected by simple relations, viz. writing Q for the gain $\frac{1}{4}k^2\sin 2\gamma$, we obtain

$$P - x = -\tfrac{1}{4}k^2\left(\tfrac{x}{y} - \tan\gamma\right)\left(\tfrac{x}{y} - \tan\alpha\right)\tan\beta,$$

which serves to find the breadths a, b, c of a given trajectory from the bifocus $PQRS\ldots$, and, if necessary, by inverting each or either as alternate ones, make the left-hand column begin with the sign $+$.

[p. 577] 26. The complete role now is for a given trajectory from the bifocus to $PQRS\ldots$, and, if necessary, by inverting each or either ones, make the left-hand column begin with the sign $+$. To find which we begin with a then if these Ω count the intersections with the alternate ones, with the left-hand column to begin with the sign $+$.

26. The complete role now $a + b + c$, and the equation of the conic surface gives $\frac{1}{2}P_1$. Operating with P_x as a tangent, then the calculation is in fact equal to the breadth in a.

27. I give some simple examples.

	0	2	4
$P = a - 1$	$-$	$+$	$-$
$Q = a - 3$	$+$	$+$	$+$
$R = a - 1$	$-$	$-$	$-$
0	P	2	Q

	0	2	4
$P = a - 3$	$-$	$-$	$+$
$Q = a - 1$	$+$	$+$	$+$
$R = a + 1$	$+$	$+$	$+$
0	P	2	Q

In the left-hand column taking the intervals to be considered $0-2$, $0-4$, $2-4$, the binbecome modified as shown are

$0-2$	$0-4$	$2-4$
$-$	$-$	$+$
$+$	$+$	$+$
$-$	$-$	$-$

viz.

Interval $0-2$, for P gain $+1$, P free; for Q gain -0, Introduction is P;

$-0-4$, $\quad -1$, $\quad \cdots\cdots = -1$; $\quad \cdots\cdots PQ$;

$-2-4$, $\quad -0$, $\quad \cdots\cdots = -1$; $\quad \cdots\cdots Q$.

And similarly in the right-hand example we have

$0-2$	$0-4$	$2-4$
$-$	$-$	$+$
$+$	$+$	$+$
$+$	$+$	$+$

[p. 578] viz. these conditions determine the unknown quantities p, R. It is at once seen that we have

$$2\theta - \beta - \gamma = \tfrac{1}{2}\pi - (a - \alpha), \quad \text{that is,} \quad 2\theta = \tfrac{1}{2}\pi - a + \alpha + \beta + \gamma,$$

and then

$$p = \Omega \sin(a - \alpha) \div \cos(\theta - \alpha) \div \sin(a - \gamma);$$

from symmetry, we see that the parabola touches also the side AB.

Suppose, next, F, G are points in the circle determined by the angles $2f$, $2g$; retaining α and θ to denote their values,

$$p = \Omega \sin(\alpha - \beta) \sin(\alpha - \gamma), \text{ and } 2\theta = \tfrac{1}{2}\pi - a + \alpha + \beta + \gamma,$$

the condition, in order that FG may be a tangent, is

$$p = \Omega \sin(\alpha - f) \sin(\alpha - g) \cos(2\theta - f - g);$$

viz. determining δ by the equation

$$a + \beta + \gamma = f + g + \delta,$$

this is

$$p = \Omega \sin(\alpha - f) \sin(\alpha - g) \cos(a\delta - \delta),$$

or, what is the same thing,

$$\sin(\alpha - a) \sin(\alpha - \beta) \sin(\alpha - \gamma) = \sin(\alpha - f) \sin(\alpha - g) \cos(\delta - \delta);$$

viz. this equation, considering therein δ as standing for $a + \beta + \gamma - f - g$, is the relation which must subsist between f and g, in order that the line FG may be a tangent to the parabola. And then, δ being determined as above, and the point H on the circle being determined by the angle 2δ, it is clear that the lines GH, HF will also be tangents to the parabola, viz. FGH will be an inscribed triangle circumscribed to the parabola; we have a single infinity of such triangles, but it is in fact, the same single infinity in form of these, f, g, satisfy only the relation $a + \beta + \gamma = f + g + \delta$, where δ has the same value but α change. The proposition is obtained as the consequence of an easily established proposition that to determine the determinations of f and α.

[p. 579] that is,

$$(b_1' - a_1')c_2 a' - (c_1' - a_1')b_2 a' + ac_1(c - b_1) = 0,$$

or, what is the same thing,

$$(b_1' - a_1')a^2 - (c_1' - a_1')b + ac_1(c - b_1) = 0.$$

Suppose, as in the figure, that P is between B and A; then, if $AF = p$, we have $a = ba + cp$, and the equation becomes

$$(b_1' - a_1')(ca + cp)^2 - (c_1' - a_1')ba + aa_1cc = 0.$$

Similarly, if, as in the figure, G is on the other side of A, that is, between A and C, and if a, a' be the dimensions AG, BG, CG, then $ba' = b'a + ca'$, that is, $cn = ba - na$, and the corresponding equation is

$$(b_1' - a_1')(ca' - na)^2 - (c_1' - a_1')ba + aa_1cc = 0.$$

We have here

$$BG\colon CF = BP \cdot CO = BC \cdot FO,$$

also,

$$pa + p'a' = AF + AF \cdot AC = FO \cdot AO = FO \cdot OH + FO \cdot a',$$

and

$$pp + p'a' = AP \cdot CF = AG \cdot CG = FO \cdot HH = FO \cdot a,$$

as may be shown without difficulty, p', a', b'' being the distances AB, BH, CH. Hence the equations become

$$c(b'_1 - a'_1) = \nu a';$$

$$b(c'_1 - a'_1) = aaa' = 0,$$

showing that, the fact being so in the figure, $b'_1 = a'_1$ and $c'_1 = a'_1$, are each of them positive, viz. that, in the geometry by the triangle $GCAU$, the radial bars a_0, a_1 are shorter than the sides b, c, respectively. Substituting for a_2, a_3, the values b, b_1, a_2 a c, also, instead of b_1, b_2, b_3, introducing the quantities b_1, a_2, b_3, where

$$b_1, b_2 = b + b_4,$$

$$b_3 = b + b_5,$$

$$b_4 = b + b_6,$$

$$b_5 = b + b_7,$$

these equations become

$$c(b'_1 - b'_2 c)ca + cba'B = a'a',$$

$$b(b'_3 - b'_4 c)ca + ba'Ca = a'a',$$

or, as these may be written, partaking the relations $B = aa + b_1 a' \sin BAC$,

$$B'\&'(b_2 - b'_4 b)c\,a + ab'\,BCH;$$

$$B'\&'(b_3 - b'_4 b)c\,a + ba'c\,AH.$$

[p. 580] All the quantities here so far being represented as positive, and the formula are applicable to the particular figures, but, to present them in a form applicable to any order of the nodes and bar, we have only to write the equations in the form

$$B(\lambda_2 - \lambda_4) = b'ca(a - x) + B(ca + b' - x)(cb - b),$$

$$B(\lambda_3 - \lambda_5) = b'ca(a - x) + B(ca + b' - x)(cc - b),$$

$$B(\lambda_4 - \lambda_4) = b'ca(a - x) + B(ca + b' - x)(cc - b),$$

and these may be replaced by the equation

$$B(\lambda_2 - \lambda_4) = b'ca(a - x) + (b' - x)(s + a),$$

$$B(\lambda_2 - \lambda_4) = b'ca(a - x) + (b' - x)(s + b),$$

$$B(\lambda_3 - \lambda_4) = b'ca(a - x) + (b' - x)(s + c),$$

since the first of these equations is implied in the other two; and then, reverting to the original form, we may write

$$BB(\lambda'_2 - \lambda_4) = BC \cdot FA \cdot GA \cdot HA,$$

$$BB(\lambda_2 - \lambda_4) = CA \cdot FB \cdot GB \cdot HB,$$

$$BB(\lambda'_3 - \lambda_4) = AB \cdot FC \cdot GC \cdot HC,$$

it being understood that the distances BC, FA, &c., which enter into these equations, are not all positive, but that they must for $\lambda \sin(\beta - a)$, $b' \sin(f - a)$, &c., and that their signs are to be taken accordingly. Or, again, these may be written

$$BC(\gamma') - \lambda_4) = FA \cdot GA \cdot HA,$$

$$CA(c_1 - c') = FB \cdot GB \cdot HB,$$

$$AB(b_1 - b_3') = FC \cdot GC \cdot HC,$$

where the signs are so just mentioned. We may say that $\frac{1}{2}(c_1 - c_1')$ is the modulus for the focus A, and the formula then shows that this modulus, taken positively, is equal to the product of the distances FA, GA, HA of A from the three nodes respectively, divided by BC, the distance of the other two foci from each other.

624. On the Bicursal Sextic

[From the Proceedings of the London Mathematical Society, vol. VII. *(1875–1876), pp. 166–172. Read March 10, 1876.]*

In the paper "On the binominal description of certain conic curves," *Proceedings of the London Mathematical Society*, vol. IV. (1873), pp. 105–111, [504], I obtained the binominal sextic as a rational transformation of a bicursal quartic. The theory was in effect as follows: taking Ω, P, Q, R, each of them a function of λ, μ of the form $(*)(\lambda,1)^3(\mu,1)^3$, and considering (λ,μ) as connected by the equation $\Omega=0$ (viz. λ, μ being coordinates, this represents a bicursal quartic), then if $\sigma=$ assume $\sim x:y:z=P:Q:R$, this is rationally connected with the bicursal quartic, viz. the points of the latter curve have a $(1,1)$ correspondence; whence the locus in question, say that curve $U=0$, is binominal. The figure is obtained as the reverse of the curve of the variable intersections of the corresponding $\lambda\mu$-curves

$$\Omega=0,\quad aP+\beta Q+\gamma R=0,$$

viz. each of these being a quartic curve having the same nodes, the nodes each count as 4 intersections, and the number of the remaining intersections is $4\cdot4-4\cdot3,=4$; and thus the curve $U=0$, $R=0$ have therefore the same 4 common intersections, that is, the curve $U=0$, $R=0$ have therefore the same 4 common intersections, then these are also 4 common intersections of the two corresponding $\lambda\mu$-curves $\Omega=0$, $aP+\beta Q+\gamma R=0$, and consequently the curve $U=0$ is binominal.

The theory assumes a different and more simple form if the second functions Ω, P, Q, R are suppose that these have no common point; the curve $U=0$ is binominal.

[p. 582] rational transformation of the cubic $\Omega=0$, is still a bicursal curve; but the order is given as the number of the variable intersections of the cubics

$$\Omega=0,\quad aP+\beta Q+\gamma R=0,$$

viz. this is $=3\cdot3-2,=7$. But if the curves $\Omega=0$, $P=0$, $Q=0$, $R=0$ have (besides the before-mentioned two common points) k other common points, then the number of the variable intersections is $=7-k$; and this is therefore the order of the curve $U=0$. In particular, if $k=1$, then the curve is a bicursal sextic. And, in the present paper, I consider the binodal sextic as thus obtained, viz. as given by the equations $\Omega=0$, $x:y:z=P:Q:R$, where $\Omega=0$, $P=0$, $Q=0$, $R=0$ are cubics, having (in all) three common points.

The bicursal sextic has in general 9 nodes; but 3 of these may unite together into a triple point: this will be the case if, in the series of curves $aP+\beta Q+\gamma R=0$, there are any two curves which have 3 common intersections with the three cubics $\Omega=0$, $P=0$, $Q=0$, $R=0$. (Observe that we distinguish thus far between the variable and the common intersections of the curves $\Omega=0$, then any linear function of P, Q, R, would have the same number of common intersections.) For, supposing the two curves to be $aP+\beta Q+\gamma R=0$ and $a'P+\beta'Q+\gamma'R=0$ and the cubic any other curve whatever

$$aP+\beta Q+\gamma R=\theta(a'P+\beta'Q+\gamma'R)$$

has the same number of variable intersections with $aP+\beta Q+\gamma R=0$, the points common to these two curves correspond on the cubic the points A_1, A_2, A_3.

The bicursal sextic has in general 9 nodes; but 3 of these may unite together into a triple point: this will be the case if, in the series of curves $aP+\beta Q+\gamma R=0$, there are any three curves which have 3 common intersections with the three cubics $\Omega=0$, $P=0$, $Q=0$. For, supposing the three curves to be

$$aP+\beta Q+\gamma R=0,\quad a'P+\beta'Q+\gamma'R=0,\quad a''P+\beta''Q+\gamma''R=0,$$

a triple point; and that to this triple point correspond the three points A_1, A_2, A_3.

We may, in the series of lines $ax+\beta y+\gamma z=0$, $a'x+\beta' y+\gamma' z=0$, rationally determine θ so that one of the three variable intersections shall correspond to A_1, A_2, or, A_2, viz. to obtain the cubic $aP+\beta Q+\gamma R-\theta(a'P+\beta' Q+\gamma' R)=0$, rationally determine θ so that one of the three variable intersections shall correspond to A_1, A_2, or, A_3, viz. to obtain the cubic to pass through one of these points A_1, A_2, A_3; the third common intersection of $aP+\beta Q+\gamma R$ and $a'P+\beta' Q+\gamma' R$ must then be on the curve; and so on, similarly. And so on, similarly. And so on, similarly. And so on, similarly. And, as before, the three points correspond to a triple point on the bicursal sextic.

[p. 583] The bicursal sextic may have a second triple point, viz. there may then exist three other common points. The theory may again be developed as before; the new triple point will correspond to a similar set of three common intersections of the cubics having the same three common points. The theory may be developed if there are six variables instead of three common intersections of the cubics having three of the variable intersections of $\Omega=0$, $aP+\beta Q+\gamma R=0$ count for a triple point; and the same set of three common intersections of $aP+\beta Q+\gamma R=0$, will count for a triple point of the second triple point. As before, we have a number of triple points such as $\Omega=0$, $a''P+\beta''Q+\gamma''R=0$, which thus form (in the same hexagonal locus) the cuspidal sextic. By varying the parameters we obtain the curve $\Omega=0$, and the points P, Q, R are the third points of intersection of the cubic with the lines BC, CA, AB respectively.

I write for greater convenience λ, ρ in place of a, μ as made by writing Ω, P, Q, R, each of them in the form $\Omega=aP+\beta Q+\gamma R$, where the points P, Q, R are the third points of intersection of the cubic with the lines BC, CA, AB respectively.

I write for greater convenience $\frac{P}{R}$, $\frac{Q}{R}$ in place of x, y, so as to make Ω, P, Q, R each of them cubic function of (λ,ρ); and I give to these functions, not the most general values belonging to a bicursal sextic with two triple points, but the values in the form obtained for them, as appearing further on, in the problem of three-bar motion: viz. the equations $\Omega=0$, $x:y:z=P:Q:R$ are respectively taken to be

$$\Omega=(a\lambda+\lambda^2\rho)(c+\rho)(a'+b\rho)(c'+a^2\rho^2)=0,$$

$$x:y:z=\lambda\rho:a'+b\rho:c'+a^2\rho^2,$$

respectively as to the form of $\Omega=0$, see the lines down in the figure.

The base curves $\Omega=0$, $P=0$, $Q=0$, $R=0$ have thus the three common intersections $(\rho=0)$, $(\rho=0)$, $(\rho=0)$, with the three points represented in the figure as the points of B, D, H in the corresponding to the common intersection $(\rho=0)$.

[p. 584] viz. excluding the fixed points A, B, C, the six intersections are C, F, G, and three intersections by the line $\Omega=0$, viz. those are the points

$$(\lambda,\rho,\mu)=(0,0,1),\quad (0,p,-f),\quad (h,0,-f),\quad (C,F,G).$$

The equation $Q+\theta R=0$ is $a'+b\rho+\theta(c+\rho\mu)=0$; viz. this is, for $\theta=0$, an equation of the second order in λ and ρ in the curve through the three points A, B, H, viz. those are the points

$$(\lambda,\rho,\mu)=(0,0,1),\quad (0,p,-f),\quad (h,0,-f),\quad (C,F,G).$$

To find the tangents at the triple point J, corresponding to the conditions $(\lambda,\rho,\mu)=(0,0,1)$, $(0,p,-f)$, $(h,0,-f)$, we have, for the values

$$\theta=\frac{c}{f},\quad \theta=\frac{c+p}{f}\cdot\frac{c+p}{f-1},\quad \text{or}\quad \theta=\frac{c}{f},\quad \theta=\frac{c+p}{f-1}.$$

Hence, considering for C as origin, the equation is

$$1+2a+\theta y=0; \quad \theta(z+y)+abz(z+y)+\theta\theta(xx+y)=0;$$

or, the equation $Q+\theta R=0$ is similarly rendered; and so, similarly, K, M at the points L, M, we have f, a, as before constraint of f, h. As tangent at the conic point.

In fact (and this is well known to be the case for a singular case of a bicursal sextic with two triple points, but in the present case they are conics, C, F, G, are independent of θ, viz. they are the points

$$(\lambda,\rho,\mu)=(0,0,1), \quad (0,p,-f), \quad (h,0,-f), \quad (C,F,G).$$

There are equivalent in virtue of the equation $Q+\theta R=0$, that

$$(\lambda,\mu)=(0,0), \quad (\lambda h,-f,0), \quad \text{(those are also the points)} \quad C,F,G,\text{etc.}$$

To deduce these expressions, writing

$$M=c+2a(\nu+p),$$

we have $K=(a+p)+c(a+b\mu)$. Hence

$$J=K+Y=(a-p)V\lambda,$$

also, from the original form,

$$J'=b+h\nu p+\lambda(C\eta+2c)-h\mu(C\eta b+Y),$$

The final values are

$$\Omega=-K^2+KX+4a\rho(M+a\mu)R+a(bRY-bzp+Y),$$

$$P=(a-\nu)\mu^2+(hR+a\mu)\Pi-4a\nu(b-Z)Y,$$

$$Q=a+KaR+Pq\lambda-a\nu Y+ba\nu c\nu),$$

$$R=b(2a+4ac\nu c+bR)$$

[p. 585] The bicursal sextic may have a second triple point, viz. there may then exist three other common points. The theory may again be developed as before, but here three other common points and the figure of the bicursal sextic. There may also be other singular cubics, see the figure of the three-cubic sextic, see the figure $\Omega=0$.

If denote like functions $\cos\theta+i\sin\theta$, $\cos\phi+i\sin\phi$ of the angles θ, ϕ, which λ, μ are subscribed to make in connecting set of angle of the bicursal sextic of the binormal cubic, Ω, C_2, R, see, taking X, Y as rectangular rectangular coordinates, x, y, and ν here their first letter for $\cos\theta+i\sin\theta$, $\cos\phi-i\sin\phi$,

$$X=a, \quad \cos\theta+b_2\sin\alpha,$$

$$X+iY=b\cos C+(\cos\phi)C_2,$$

$$X-iY=b\cos C-(\cos\phi)C_2,$$

viz. the first two intersect in the point $(0,0)$, the second two in the point $(b\cos C,\ b\sin C)$, the first and third of these on the curve at the point $X=b\cos C$, $b\sin C$ (with from the same point of b, forming, with B and C, a triad of foci.

Verification of Ω.—The equation is

$$\Omega=(a\lambda+\lambda^2\rho)(c+\rho)(a'+b\rho)(c'+a^2\rho^2)=0,$$

where
$$\Omega = (a\lambda + \lambda^2\rho)(c+\rho)(a' + b\rho)(c' + a^2\rho^2) = 0,$$
or, putting for shortness
$$\lambda = a + 4a + b\mu \quad \mu' \quad \nu' = (X - iY)(b\cos C + i\sin C) + \cos C - i\sin\theta - \nu' + \nu''(a' + \mu\nu),$$
$$M = b\nu + (X - iY)V\lambda + b\nu + ac\nu - a\nu + p\nu - \nu + \nu'\nu' + ac\nu = 0,$$
or, the trial may be made, a, ν being constants, the values $c\nu$ and the dual function K, X, Y.

[p. 586] Writing for homogeneity $\frac{x}{z}$, $\frac{y}{z}$ in place of λ, μ, and $\frac{x}{w}$, $\frac{y}{w}$ in place of x, y, the equations become
$$(ax^3 + a^4\rho^2) + a\rho(b + b_4b\rho)(b + b_4\rho)(c^2 + b_4\rho^2) = 0,$$
and
$$x : y : z : w = \lambda(\lambda + b\rho\mu) : \left(\frac{b}{c} + acb\mu\right) c' : c\rho\nu.$$

Comparing with the foregoing equations
$$x\lambda\rho + f(\lambda^2 + a)\nu = g(c^2 + a)\mu + ac\nu + bu(\lambda^2 + \nu)$$
and
$$x : y : z = \lambda\nu : (a' + b\rho) : (c' + a^2\rho^2),$$
the equations agree together, and we have
$$\rho = a\mu(c' + a)(\mu')z',$$
$$\mu = ecc\nu,$$
$$\Omega = cc\nu(\nu + 1)\nu(bu),$$
$$\beta = cc\nu(\nu + 1)\nu(bu);$$
the integrals at the triple points then are
$$\nu = 0, \qquad \rho = 0,$$
$$a\mu, c\mu(\nu, c) = 0, \qquad a\mu - ac\mu, c = 0,$$
$$\alpha = c\mu(c\mu),$$
$$b\mu = 0, \qquad b\mu = (\nu, c) = 0,$$
viz. writing the rectangular coordinates, and for y substituting the value $\cos C + i\sin C$,
$$X + iY = 0, \qquad X - iY = 0,$$
$$X + iY = b\cos C + (\cos\phi)C_2, \qquad X - iY = b\cos C - (\cos\phi)C_2,$$
$$X + iY = \nu,$$
viz. the first two intersect in the point $(0, 0)$, the second two in the point $(b\cos C, b\sin C)$, and the third pair of three of the points of the figure, viz. the focuses with A, B and C, a triad of foci.

[p. 587] (omitted as overlap)

[p. 588] (continuation)

[p. 589] (continuation)

625. On the Condition for the Existence of a Surface Cutting at Right Angles a Given Set of Lines

[From the Proceedings of the London Mathematical Society, vol. VIII. *(1876–1877), pp. 53–57. Read December 14, 1876.]*

In a congruency or doubly infinite system of right lines, the direction-cosines a, β, γ of the line through any given point (x, y, z), are expressible as functions of x, y, z; and it was shown by Sir W. R. Hamilton in a very elegant manner that, in order to the existence of a surface (or, what is the same thing, a set of parallel surfaces) cutting the lines at right angles, $a\,dx + \beta\,dy + \gamma\,dz$ must be an exact differential: when this is so, writing $V \int(a\,dx + \beta\,dy + \gamma\,dz) = 0$, the equation of the system of parallel surfaces each cutting the given lines at right angles.

The proof is as follows:—If the surface cuts, its differential equation is $a\,dx + \beta\,dy + \gamma\,dz = 0$, and this equation must therefore be integrable by a factor. Now the functions a, β, γ are such that $a^2 + \beta^2 + \gamma^2 = 1$, and they hereby satisfy a system of partial differential equations which Hamilton deduces from the geometrical notion of a congruency; viz. posting from the point (x, y, z) to the consecutive point on the line, that is, to the point whose coordinates are $x + ap$, $y + \beta p$, $z + \gamma p$ (p infinitesimal), the line through this point is parallel to the original line; and consequently the direction-cosines of this line are unchanged by the variations $\Delta x = ap$, $\Delta y = \beta p$, $\Delta z = \gamma p$, respectively. We thus obtain the equations

$$a\frac{da}{dx} + \beta\frac{da}{dy} + \gamma\frac{da}{dz} = 0,$$

$$a\frac{d\beta}{dx} + \beta\frac{d\beta}{dy} + \gamma\frac{d\beta}{dz} = 0,$$

$$a\frac{d\gamma}{dx} + \beta\frac{d\gamma}{dy} + \gamma\frac{d\gamma}{dz} = 0.$$

[p. 591] so that the required equation is a, B, γ will be satisfied if only

$$\frac{da}{dx} + \frac{d\beta}{dy} + \frac{d\gamma}{dz} = 0,$$

and it is to be shown that these conditions hold good.

Writing for a, B, γ the below-mentioned values, we find

$$\frac{da}{dx} = \frac{dC}{dx}\frac{1}{a+\lambda}, \qquad \frac{d\beta}{dy} = \frac{dC}{dy}\frac{1}{a+\mu}, \qquad \frac{d\gamma}{dz} = \frac{dC}{dz}\frac{1}{a+\nu},$$

and it is to be shown that these conditions hold good.

Writing for a, B, γ the below-mentioned values, we have

$$\frac{da}{dx} = \frac{dC}{dx}\cdot a\frac{a+\lambda}{a+\mu} + \frac{d}{dx}\left(\frac{a}{a+\lambda}\right)C = \frac{a^2}{a+\lambda}\frac{dx}{dx} + C^3\frac{dx}{dx}\left(\frac{a}{a+\lambda}\right),$$

and similarly for $\frac{d\beta}{dy}$ and $\frac{d\gamma}{dz}$ such that

$$\frac{da}{dx} + \frac{d\beta}{dy} + \frac{d\gamma}{dz} = 0, \qquad \text{viz.} \quad \frac{da}{dx} + \frac{d\beta}{dy} + \frac{d\gamma}{dz} = 0,$$

and thence without difficulty

$$\left(\frac{a^2}{a+\lambda}\frac{da}{dx} + \frac{a^2}{a+\mu}\frac{d\mu}{dx}\right) - \left(\frac{1}{1+d\beta}\frac{d\beta}{dx} + \frac{d\nu}{dx}\right) - \omega = \left(a\frac{dx}{dx} + b\frac{dy}{dy} + c\frac{dz}{dz}\right),$$

$$\left(\frac{a^2}{a+\lambda}\frac{d\lambda}{dy}+\frac{d\mu}{dy}\right)+\frac{1}{2(a+\lambda)}=-m\,a\;(\quad)+(\quad)\,,$$

$$\left(\frac{a^2}{a+\lambda}\frac{d\lambda}{dz}+\frac{a^2}{a+\mu}\right)\frac{dz}{dz}=-\frac{1}{2}\quad(-)+(-).$$

626. On the General Differential Equation $\dfrac{dx}{\sqrt{X}} + \dfrac{dy}{\sqrt{Y}} = 0$, where X, Y are the same Quartic Functions of x, y respectively

[From the Proceedings of the London Mathematical Society, vol. VIII. (1876–1877), pp. 184–199. Read February 8, 1877.]

[p. 592] WHERE $\Theta = a + b\theta + c\theta^2 + d\theta^3 + e\theta^4$, the general quartic function of θ, and let it be required to integrate by Abel's theorem the differential equation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0.$$

We have

$$\begin{vmatrix} x^4, & x, & 1, & \sqrt{X} \\ y^4, & y, & 1, & \sqrt{Y} \\ z^4, & z, & 1, & \sqrt{Z} \\ w^4, & w, & 1, & \sqrt{W} \end{vmatrix} = 0,$$

a particular integral of the above equation, taking therein z, w as constants, is the general integral of

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0,$$

viz. the two constants z, w enter in such wise that the equation contains only a single constant; whence also, attributing to w any special value, we have the general integral with z as the arbitrary constant.

[p. 593] Take $w = x$; the equation becomes

$$\begin{vmatrix} x^4, & x, & 1, & \sqrt{X} \\ y^4, & y, & 1, & \sqrt{Y} \\ z^4, & z, & 1, & \sqrt{Z} \\ 1, & 0, & 0, & \sqrt{W} \end{vmatrix} = 0,$$

a relation between x, y, z which may be otherwise expressed by means of the identity

$$x(P + \beta\theta + \gamma\theta^2) = (a\theta + \beta) - (a\theta + \beta)\theta(\theta - z)(\theta - z),$$

or, what is the same thing,

$$x(2x + \beta) - z = -(2(a\theta - \theta)(\theta + \theta) - z),$$

$$+2\beta y = -a + (3z - d) - (a + xy + z),$$

where β, γ are indeterminate coefficients which are to be eliminated. Write

$$x^2 = \frac{\sqrt{X}}{xz} = P, \quad y^2 = \frac{\sqrt{Y}}{yz} = Q;$$

then we have $\beta + \gamma = P + Q$, $\beta y + \gamma + Q = z$; giving

$$B + \gamma = 1 + Q - P, \quad B\gamma + Q = z - z.$$

Substituting these values in the first of the preceding three equations, we have

$$x\frac{2(Py - Qx)(x - y) + (Q - P)y}{(x - y)^2} - x = \frac{2 + (Q - P)y - Q}{(x - y)}\,(ay - z + s)\,(x + y + z),$$

that is,
$$\frac{2(Qy-Px)}{x-y}=\frac{2(2+(Q-P)y-Q)}{x-y},\quad (x+y+z)\cdot\frac{2}{x-y}\cdot\frac{1}{x}\cdot\frac{1}{y};$$
or, reducing by
$$Qy-Py-x=a\frac{x-y\sqrt{Y}}{x-y\sqrt{X}}.$$

[p. 594] times, the rational part is
$$M'=c+d(x+y)+e(x^2+xy+y^2),$$
and the two terms involving the operator L are as before stated that, in the particular case where $e=0$, the first equation becomes
$$M'=e+d(x+y)+e(x^2+xy+y^2),$$
and the two results for this case may be perfectly stated $C-e+0dt$.

But it is required to identify the two solutions in the general case where a is not $=0$, I remark that I have, by my *Treatise on Elliptic Functions*, Chap. XIV., further developed the theory of Euler's relation, and have shown there, that, regarding C as variable, and writing
$$\Omega=ax^2+2xe(x+y)e^{-y}+4z(xx^2+2z^2x+4zy)(xe-ry)ee+2xye(x+y)e^{-y}+\frac{1}{2}y(xe+xy+xy+e)(C-exy),$$
then the equation between the variables x, y, C corresponds to the differential equation
$$\frac{dx}{\sqrt{X}}+\frac{dy}{\sqrt{Y}}=\frac{dC}{\sqrt{\Omega}}=0,$$
a result which will be useful for effecting the identification. The Abelian solution may be written
$$P\left[\frac{1(xy(\sqrt{X}+y\sqrt{Y})}{(x-y)^2}-x^2\right]=\frac{1}{x-y}\sqrt{\Omega}+4ex(x+y)e^{-y}-4ex(c+ey)=2EX',$$
and substituting for M its value, and multiplying by $(x-y)$, the equation becomes
$$\begin{aligned}&2\sqrt{x}(x-y)(z+X+y\sqrt{Y})(P+y\sqrt{P+Q}+(Q'(x-y)\sqrt{Y}))\\&=2(c+d+(x+y)+e(x^2+xy+y^2)e+xze^{-y}2(e\sqrt{X}-x\sqrt{Y});\end{aligned}$$
and substituting therein for M its value, and multiplying by $(x-y)$, the equation becomes
$$\begin{aligned}&2x((x+y)(\sqrt{X}-y\sqrt{Y})+2x(x+y)((c+ey)-(x+y)\sqrt{Y}-x\sqrt{Y}))\\&=2(c+d(x+y)+e(x^2+xy+y^2))+xz-2(c+ey-y)e^{-y}2(x\sqrt{Y}-x\sqrt{Y});\end{aligned}$$
On the left-hand side, the rational part is
$$2X+Y+e(-y)(x^4+2xy+y)+4e^{-y}(c-e^{-y}2(x+y)e^{-y})y+2(c+d)(x-y)(x+y),$$
which, substituting therein for X, Y their values, becomes
$$=2a+b(x+y)+c(x^2+4xy+y^2)+2dxy(x+y)+2ex^2y^2((x+y),$$
and the irrational part is at once found to be
$$=2(a+(a-y)\sqrt{Y}-y(a+(a-y))\sqrt{X}).$$

[p. 595] Multiplying the numerator and the denominator by

$$d(x-y)+3e(x^2-y^2)+2e\sqrt{x}((y-1)(X-y^4Y)),$$

the denominator becomes

$$=(x-y)^2\left[d+3e(x+y)+4e^{-y}(\sqrt{x}-1)\sqrt{X}\right],$$

which, introducing therein the C of Euler's equation, is

$$=(x-y)^2(C-6acC).$$

We have therefore

$$x(x-y)-6ac=(x(x-y)C+b(x-y)+2ay(P+2dx(yP-xQ)+2xx^2(\sqrt{Y}P+y\sqrt{X}Q),$$

$$+2xy(xx-y)\sqrt{X}-x\sqrt{Y}\cdot Vx((y-1)(X-y\sqrt{Y})((x-y)(x\cdot s-2))-2(x-y)\sqrt{Y\sqrt{Y}}.$$

Using b, to denote the same value as before, the function on the right-hand is, in fact,

$$=(x-y)^2\left[Bx-ad+de\sqrt{x}\sqrt{Y}\right],$$

and, this being so, the required relation between x, C is

$$x(\beta x-4ac)=(Bx-ad+de\sqrt{x}\sqrt{Y}),$$

$$((x-y)\sqrt{Y-y\sqrt{X}})((x-y)\sqrt{X}\cdot s-2)).$$

To prove this, we have first, from the algebric

$$\left(\frac{\sqrt{X}-\sqrt{Y}}{x-y}\right)^2=C+d(x+y)+e\sqrt{x}\sqrt{X},$$

to express Φ as a function of C as a variable, gives

$$\frac{dx}{\sqrt{X}}+\frac{dy}{\sqrt{Y}}+\frac{dC}{\sqrt{\Omega}}=0,$$

and we have therefore

$$-\sqrt{\Omega}=\sqrt{x}\frac{dC}{dx}\quad(=-\sqrt{Y}\frac{dC}{dy}),$$

viz. $a\sqrt{x}\frac{dC}{dx}$ will be a symmetrical function of x, y. Putting, as before

$$M=\frac{C-\sqrt{Y}}{x-y}.$$

we have

$$C=M^2-d(x+y)+e(x+y),$$

and thence

$$\Omega=M^2-x(x+y)+e(x+y)^2.$$

We have

$$\frac{dM}{dx}=\frac{1}{x-y}\left(\frac{X'}{2\sqrt{X}}-\frac{\sqrt{X}-\sqrt{Y}}{x-y}\right),$$

[p. 596] and hence

$$\sqrt{\Omega} = \sqrt{X} - \sqrt{Y}\left[2M\frac{dM}{dx} - d - 2e(x+y)\right],$$

which thus is $(x-y)$, the numerator becomes

$$M' = c + d(x+y) + e(x^2+xy+y^2),$$

say that the particular case as may putting $C = c + exy$.

But it is required to identify the two solutions in the general case where e is not $= 0$. I remark that I have, by my *Treatise on Elliptic Functions*, Chap. XIV., further developed the theory of Euler's relation, and have shown there, that, regarding C as variable, and writing

$$\Omega = \frac{(C-c)^2(C-4ac)}{C(\theta-y)} - 4a\left[b - c(x+y) + d(xx+y^2) - 2e(xx+xy+y^2)\right](x-y)^2\sqrt{X}\sqrt{Y};$$

and substituting therein for M its value, and multiplying by $(x-y)$, this is

$$\Omega = c(P-p)[2P - 4ac + 2(x-y)^2\sqrt{X}\sqrt{Y}].$$

To prove this, we have first, from the algebric

$$\sqrt{\Omega} = c(P-p)\sqrt{\Omega(P-p)\sqrt{X}\sqrt{Y}},$$

to express Φ as a function of C as a variable. This equation, regarding C as a variable, gives

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dC}{\sqrt{\Omega}} = 0,$$

and we have therefore

$$-\sqrt{\Omega} = \sqrt{x}\frac{dC}{dx} \quad (= -\sqrt{Y}\frac{dC}{dy}),$$

viz. $\sqrt{x}\frac{dC}{dx}$ will be a symmetrical function of x, y. Putting, as before,

$$M = \frac{C - \sqrt{Y}}{x-y},$$

we have

$$C = M^2 - d(x+y) + e(x+y)^2,$$

and thence

$$\frac{dC}{dx} = 2M\frac{dM}{dx} - d - 2e(x+y),$$

that is,

$$\sqrt{\Omega} = \sqrt{x}\left[2M\frac{dM}{dx} - d - 2e(x+y)\right] \quad \text{suppose;}$$

or, writing for shortness X', M' to denote the derived functions $\frac{dX}{dx}$, $\frac{dM}{dx}$, (Y' is afterwards written to denote $\frac{dY}{dy}$), but as the final formula contains only X', $\frac{dX}{dx}$, and Y', $\frac{dY}{dy}$. (This does not assume any defect of symmetry), we find

$$\Omega = N\left[BC - X(X - 2(x+y) + \sqrt{X}\sqrt{Y}\right]$$

$$= D^2\sqrt{X^2} + 2(xx-y)ex^2\sqrt{X^2} + 2(c-0)\sqrt{Y\sqrt{XY}} - X'\sqrt{Y}],$$

[p. 597] viz. this is

$$= [2a + b(x + y) + c\sqrt{x}(y - z) + e(x + y)(\sqrt{Y} - x\sqrt{X}))]$$
$$\times[d(x - y) + 2e(xx + 2xy + yx)],$$
$$+\frac{1}{2}\sqrt{X}(d(x - y) - 4e(xx + 2xy + y^2) - 2exex(x - y + (z + y)e^{-y}((x - y)x + y\sqrt{Y})),$$

which is, in fact, equal to the expression on the left-hand side.

To complete, then, it is required to express $\sqrt{X}$ as a function of x, y symbolic that

$$x(x - y)(\sqrt{X} - \sqrt{Y}) \cdot 2(x + y)x \cdot 4 = (x - y)(\sqrt{X} - \sqrt{Y}) \cdot 2(x - y)(\sqrt{X} + \sqrt{Y}),$$

which may be done by putting therein

$$2P = -X^2X + 2xexy(X - y) + 4ex(\sqrt{x}y - \sqrt{y}x),$$

which when expanded into X as a function of x, y, on X as a function of x, y, of which there are forms in x, y, on $X = 0$, in fact, equal to the corresponding expression on the left-hand side.

So the left-hand side, considering therein down as they immediately present themselves; but, of course, the relation X is $X(x+y)-$ to come $a+(x+y)-$, when these are written in the form

$$-b + dxy = (a + b),$$

which is right.

To X verification of Ω.—The quantity is

$$J\Omega(X - y)X + 8x(c - y)\sqrt{C}(C - 4ac) + 4a(x - y)\sqrt{X}\sqrt{Y}$$
$$= 2L(P - xP\sqrt{X}) - 4a(P\sqrt{X}\sqrt{Y} - 2yP + \sqrt{Y}(c - 4ac))((x - y)\sqrt{Y} - y\sqrt{X})$$
$$+2\sqrt{X}((x - y)\sqrt{Y} - x\sqrt{Y})((x - y)\sqrt{X}(c - 1)) = 0;$$

which, throwing out the factor $x - y$, becomes

$$0 = 2\sqrt{X}(c-y)+2cxy(x-y)+2cxxy-2(c+y)e\sqrt{X}\sqrt{Y}-4\sqrt{X}-2(x+y)-2cxy-2cxy-2(x+y)y^2,$$

which, throwing out the factor $x - y$, is

$$0 = 2N\left(BC\sqrt{X} - 2(x + y)yP + 2cxy\sqrt{X}\right) + 2(c - 4ac)\sqrt{Y}((x - y)\sqrt{Y} - y\sqrt{X}),$$
$$+2\sqrt{X}((x - y)\sqrt{X} - x\sqrt{Y})((x - y)\sqrt{X}(c - 1)).$$

[p. 598] and similarly herein for X, Y their values, and arranging the terms, we find

$$\Omega = a\sqrt{X}\sqrt{Y}\sqrt{X}((x - y)\sqrt{Y} - y\sqrt{X}),$$

where

$$\Omega = a[2X + (a - y)X]$$
$$-2(x - y)yX$$
$$-(xy + (a + y)y)$$
$$-(x - y)(ay - y)$$
$$+By(2X + (x - y)X)$$
$$+2Y(x - y)X,$$

$$\Omega = 4xy(a-y)X$$
$$-2(a-y)y(2X+(x-y)X)$$
$$-4XY$$
$$+2by(x-y)X$$
$$+4cx(x-y)X$$
$$+4ex(x-y)(y-2)X,$$
$$\Omega = -4xy(a-y)X$$
$$+2cx(2X+(a-y)X)$$
$$+4xY$$
$$-2cx(x-y)X$$
$$-4x(x-y)X$$
$$+4x(x-y)(y-2)X,$$

where the same identifications hold down with the formulae above stated, refusing, except differently and averaging, we have here so much.

To reduce these expressions, writing $M = 2x(x+y)$, hence we have

$$J' = 2x + 4xy + e^{-y}((y+y)y + 2xy + Y), \quad 2xy(c+y)(y+Y+Y),$$

which divides by $(x-y)$.

Hence the equation now is

$$(x-y)^2\nabla = (a+b+e)(\sqrt{Y}-y)\cdot x\cdot\nabla((ax-y)y\sqrt{Y}),$$
$$+y(2xx+y) - x(b+e)(a+4ay-a)(c\sqrt{X}-1) + 2x((x-y)x - y\sqrt{Y}-x);$$
$$+ay\nabla(b+e)((x-y)x - y\sqrt{Y} - 4y\sqrt{X})\cdot 2(x-y).$$

The function on the right-hand is, in fact,

$$= 2(L\nabla + (x+y))\cdot ax\cdot\nabla\cdot\sqrt{Y} - y\sqrt{X},$$

and the irrational part is at once found to be

$$= 2(a+(a-y)\sqrt{X} - y(a+(a-y))\sqrt{X}).$$

[p. 599] *Verification of* Ω.—The equation is

$$-X' + 4\nabla Y + (x-y)(X' + (\sqrt{X}-4)y\sqrt{X})M + xyy,$$
$$= 2L(M'\cdot\sqrt{X}-y) + 4xx\sqrt{X^2}M + 2N\sqrt{X^2}M;$$

or, putting therein

$$\frac{X'}{(x-y)} = X' + 8X' - 6\sqrt{X}((\sqrt{X}-y)P)M^2 = 2PxM.$$

this is

$$(x-y)^2 = X' + 4XY$$
$$+2X'\cdot X - 8a(x-y)xP$$

$$+(a-y)(2xx+y)-4(\sqrt{X}\cdot 2(y-y))x\sqrt{X}M,$$
$$-2(\sqrt{X}-y)((x-y)X-xy^2\sqrt{X})-4(c-xy)(c-xy^2)\cdot 2\sqrt{X}\sqrt{Y};$$
$$+(x-y)(2X+4y(c\sqrt{X}+y)P+2(cxy\sqrt{Y}\cdot 2X'+2y\sqrt{X})X^2),$$
$$+(x-y)\cdot\sqrt{X}(c+y)X\cdot xz\cdot 2(X'\cdot x+xy\sqrt{X})-2y\sqrt{X}.$$

We have $YE'F=x(X+1)+de\sqrt{x}(X+Y-4yx+2by+ex^2)$ and thence

$$-X'\cdot 2X+(a-y)y(X'\cdot 4\cdot c\cdot 2\cdot cX)x\sqrt{X}+2\sqrt{X}(c\cdot c\sqrt{X}\sqrt{Y}-((x-y)-x)y\sqrt{X}),$$

or, as this may be written,

$$\Phi=-X'+2\sqrt{X}P.$$

[p. 600] We find, moreover,

$$\Phi'=X'-4(c+y)x+(x-y)((\sqrt{X}-4)y\sqrt{X})Xy\sqrt{X},$$

which divides by $(x-y)$, viz. this is

$$=y(c+(a-y)y)+4y(a+(a-y))+2X+2y(x-y)yX,$$

which divides by $(x-y)$, viz. this is

$$=y(c+a(x+y))\sqrt{Y}-y(b+2ay+2c)\cdot 2y-y\sqrt{X}.$$

Hence above the values reducing, and throwing out the factor X, the equation becomes

$$-b+d\,ay=(a+b),$$

which is right.

To the first line, in the equation immediately preceding, we have

$$J'=-2(x-y)x\nabla x\cdot c=-2(a-y)\sqrt{X}-xPx\nabla cy\sqrt{X};$$

which is

$$=-2X+(a-y)Py\cdot xc+(b+e)((x-y)X-y\sqrt{X}P)x-x;$$

hence, throwing out the factor $(x-y)$,

$$\Phi=aPx\nabla c-x((x-y)X-y\sqrt{X}P)-(a+y)Py\cdot 2xc-2xPx\nabla,$$

which is to come $(x-y)P\sqrt{X}P+2P\sqrt{X}$,

$$0=2XY+2(a+b)x+(x-y)P\nabla+2X\nabla,$$

which, throwing the factor $(x-y)$, is

$$0=-X'+2Xy\nabla+(x-y)\cdot c\cdot cxy\sqrt{X},$$

which, throwing the factor $(x-y)$, becomes

$$0=-X+2\sqrt{X}P,$$

which is the required relation.